
VVUQ Pipeline Engineering Notebook

project: vvuv-pipeline | page: 1 | seed: 20260525

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Index terms

Scope of this notebook

This notebook records the day-to-day engineering of the VVUQ pipeline that holds verification and uncertainty quantification ahead of code generation. Entries are append-only and dated; numbers are reproducible from the recorded seed.

Log entries

Entries are append-only; each carries a date and an outcome.

2026-05-25 wired the ten-gate surface; first dry run held the verification fraction at 1.0 over 51 candidates

2026-05-26 added the coefficient-of-variation bound per gate; three catastrophe gates pinned at agreement 1.00

2026-05-27 32-iteration sweep clears; composite mean 93.56; archived the run index

Figure placeholder (Images/pipeline.png). The float, caption, and
\\label mirror the VVUQ-02 figures; replace the box with
\\includegraphics.

Figure 1: Figure 1. Placeholder schematic of the pipeline stages exercised in the log.

Run summary

Run	Inputs	Result
dry-run	51 candidates	51/51; 1 ACCEPT, 5 BLOCK, 1 ESCALATE
sweep	32 iterations	32/32 clear; mean 93.56

Table 1: Table 1. Run summary (modeled on VVUQ-05 Table 1).

Observations

The pipeline is bottlenecked by validation-agreement evaluation, not verification; future work should parallelize the agreement step without weakening any gate bound.

Acknowledgments

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Cite This Manual

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Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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